FIITJEE Students From All Programs have bagged 34 in Top 100, 66 in Top 200 and 174 in Top 500 All India Ranks. FIITJEE Performance in JEE (Advanced), 2014:
2521 FIITJEE Students from Classroom / Integrated School Programs & 3579 FIITJEE Students from All Programs have qualified in JEE (Advanced), 2014.

ALL INDIA TEST SERIES

FIITJEE

JEE(Advanced)-2015

ANSWERS, HINTS & SOLUTIONS

FULL TEST– IV

(Paper-2)

Q. No.	PHYSICS	CHEMISTRY	MATHEMATICS
1.	**D**	**A**	**A**
2.	**C**	**B**	**B**
3.	**D**	**D**	**C**
4.	**B**	**A**	**D**
5.	**A**	**A**	**B**
6.	**C**	**D**	**B**
7.	**A, D**	**A, B, C**	**A, C, D**
8.	**B, C, D**	**A, B, D**	**A, B, C**
9.	**A, D**	**A, C, D**	**A, B, C, D**
10.	**C**	**B**	**B**
11.	**C**	**A**	**C**
12.	**A**	**C**	**B**
13.	**A**	**B**	**A**
14.	**B**	**C**	**C**
1.	**A → p** **B → q** **C → r** **D → s**	**A → (p, q, r)** **B → (p, q, r)** **C → (p, s)** **D → (p, r, t)**	**(A) → (r)** **(B) → (p)** **(C) → (t)** **(D) → (q)**
2.	**A → q** **B → p** **C → p** **D → q**	**A → (p)** **B → (t)** **C → (s)** **D → (r)**	**(A) → (s)** **(B) → (p)** **(C) → (q)** **(D) → (r)**
1.	8	**5**	**0**
2.	9	**5**	**9**
3.	1	**6**	**3**
4.	3	**1**	**0**
5.	5	**4**	**1**

Physics

PART – I

SECTION – A

1. There are 8 points in space and every pair connected through C. Without loss of generality we can choose 6 of them exactly at the mid-point of the remaining two: (1, 2, 3, 4, 5 and 6 are at same potential).

   Acceleration $C_{eff} = \frac{C}{2} \times 6 + C = 4C$.

2. For zero P.D. across galvanometer.

   $i_1R_1 = i_2R_2$

   and $L_1 \frac{di_1}{dt} = L_2 \frac{di_2}{dt}$

   Now, $i_2 \propto di_2$

   and $i_1 \propto di_1$

   $\Rightarrow \frac{L_1}{L_2} = \frac{R_1}{R_2} \Rightarrow \frac{3}{L} = \frac{2}{4}$

   or L = 6 mH.

3. For any system of particles,

   $KE = \frac{1}{2}MV_{CM}^2 + \frac{1}{2}\sum M_i V_{iCM}^2$

   First term of RHS is frame dependent, second term is not frame dependent.
   Loss in KE is any frame is same as loss of KE in C.M. frame. The first term doesn't change during collision.

4. $I' = I \cos^2(D\phi/2)$

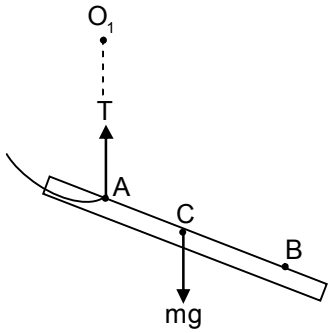
   $D\phi = \frac{2\pi}{\lambda}.Dx = \frac{2\pi}{\lambda}.\frac{dy}{D} = \frac{2\pi y}{\beta} = \frac{2\pi}{\beta}.\frac{\beta}{4} = \frac{\pi}{2}$

   $\Rightarrow I' = I\cos^2\left(\frac{\pi}{4}\right) = \frac{I}{2}$

5. Charge –Q is uniformly distributed at inner surface of cavity and +Q is uniformly distributed at outer surface. Outside energy is same as shell.

6. For a point source wave front is always spherical, after passing through lens shape becomes plane in case of convex lens as object lies on focus, but in case of concave lens it remains spherical with greater curvature.

7. FBD of rod just after the string is burnt is shown. Net force is vertical so the acceleration is vertical. Motion of point A is circular. Initial angular velocity is zero thus the radial acceleration which is vertical is zero.



8. Force of electric field is always along x-axis but force of magnetic field is always in y-z plane. Speed in y-z plane remains constant as magnetic field does not change speed.

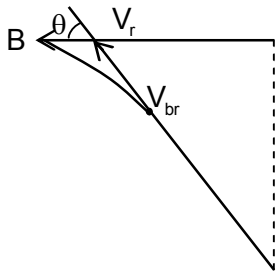
9. $k_{max} = hv - \phi$

$k'_{max} = \dfrac{hv}{2} - \phi$

electrons are ejected only if $\dfrac{hv}{2} > \phi$

and $k'_{max} = \dfrac{k_{max}}{2} - \dfrac{\phi}{2} < \dfrac{k_{max}}{2}$

10. Considering the velocities in river frame, point B appears to move left with a velocity V_r while boat has a velocity V_{br} always towards A.



Velocity of approach = $V_{br} - V_r \cos\theta$

$\Rightarrow \int_0^T (V_{br} - V_r \cos\theta)\,dt = W$ .....(1)

Also:

$V_r T = \int_0^T V_{br} \cos\theta\, dt$ .....(2)

($\because$ equating x-coordinate).

From (1) and (2):

$\Rightarrow V_{br} T - \dfrac{V_r^2}{V_{br}} T = W$

or $T = \dfrac{WV_{br}}{V_{br}^2 - V_r^2}$

11. As time passes, θ decreases, initial θ is 90°. The net speed is given by:

$V_b = \sqrt{V_{br}^2 + V_r^2 + 2V_r V_{br} \cos(180° - \theta)}$

$= \sqrt{V_{br}^2 + V_r^2 - 2V_r V_{br} \cos\theta}$.

12-14. [A and B are always at same temperature as piston is diathermic].

For whole gas system ΔQ = 0, ΔW = –Ve(as work done by gas in right is more –ve than +ve work done by gas in left) $\Rightarrow$ ΔU = +Ve

SECTION – B

1. C.M. of the cube comes down through a height of $\sqrt{3}$ m while C.M. of displaced water goes up by a height of $\dfrac{\sqrt{3}}{2}$ m.

$\Rightarrow$ Loss in PE of cube = 2000 × 10 × $\sqrt{3}$ Joule.

$= 2\sqrt{3} \times 10^4$ Joule.

Gain in PE of water = $1000 \times 10 \times \dfrac{\sqrt{3}}{2} = \dfrac{\sqrt{3}}{2} \times 10^4$ Joule.

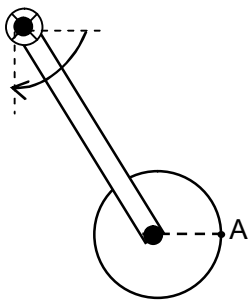
Using work energy theorem:

$W_T + W_{Gravity} = 0$

or $W_T = -W_{Gravity} = \Delta PE = \left(\frac{\sqrt{3}}{2} - 2\sqrt{3}\right)\times 10^4 = \frac{-3\sqrt{3}}{2}\times 10^4$ Joule.

SECTION – C

2. Motion of disc is only a pure translational motion, as torque on disc about C is always zero, even if C is accelerated. When rod becomes vertical, A remains horizontally to the right of C. Velocity of whole disc is same as C. Net velocity of disc be V.



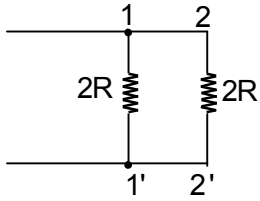
Loss in potential energy = Gain in kinetic energy

$$mg\left(\frac{3r}{2}\right) + mg3r = \frac{1}{2}\left[\frac{m(3r)^2}{3}\right]\left(\frac{V}{3r}\right)^2 + \frac{1}{2}mV^2$$

$$\Rightarrow V = \sqrt{\frac{27}{4}gr} = 9\,\text{m/s}.$$

3. $V_{22'} = \eta V_{11'}$
$V_{33'} = \eta V_{22'} = \eta^2 V_{11'}$
So on, $V_{44'} = \eta^3 V_{11'}$

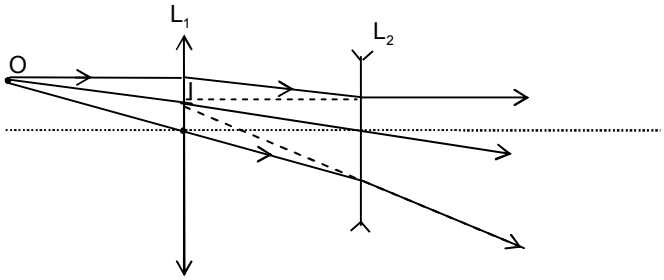
$$\Rightarrow 2 = \eta^3(16) \Rightarrow \eta = \frac{1}{2}$$



It means potential get equally divided in consecutive sections.
The equivalent arrangement is

4. After crossing L_1, rays become parallel and crossing L_2 the rays diverge to create image point at it's focal plane which is x = 0. Ray crossing optical centre of L_2 passes undeviated and y coordinate of this ray is focal plane of L_2 gives y coordinate of image, which is

$$\frac{1}{2}(6\text{mm}) = 3\text{mm}.$$



5. $$\vec{r} = A\left(\frac{1+\cos^2\omega t}{2}\right)\hat{i} + B\left(\frac{1-\cos^2\omega t}{2}\right)\hat{j}$$

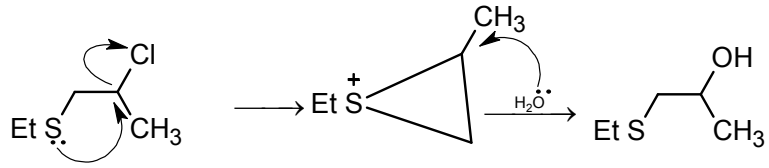
$$= \frac{A}{2}\hat{i} + \frac{B}{2}\hat{j} + \left(\frac{A}{2}\hat{i} - \frac{B}{2}\hat{j}\right)\cos\omega t = \vec{r}_0 + \vec{\alpha}\sin(\omega t + \frac{A}{2})$$

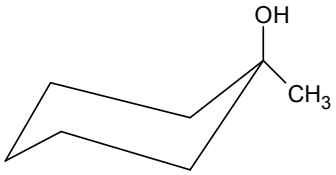
Amplitude is $|\vec{\alpha}| = \frac{\sqrt{A^2+B^2}}{2} = 5$.

Chemistry

PART – II

SECTION – A

1. 

2. 

Give 3° carbocation as intermediate while other gives 2° carbocation.

4. $KI + I_2 \longrightarrow KI_3$

6. $r = \frac{0.529 \times n^2}{1} A^o$

$\therefore$ 0.529 $[(n + 1)^2 - n^2] = 0.529 (n - 1)^2$

on solving, n = 4

13. $\kappa = \Lambda_m.C = \frac{\Lambda_m \times M}{1000} = \frac{200 \times 0.04}{1000}$

$= 8 \times 10^{-3}$ S cm^{-1}

$\kappa = G\left(\frac{\ell}{A}\right) \Rightarrow 8 \times 10^{-3} = G\left(\frac{0.50}{2}\right)$

G = 0.032 S, V = IR $\Rightarrow$ I/G

I = 5 $\times$ 0.032 = 0.16 A

SECTION –C

1. $d(kE) = mv.dv = mv\frac{h}{4\pi m\Delta x}$

$= \frac{3 \times 10^8}{3} \times \frac{6.62 \times 10^{-34}}{4 \times \pi \times \frac{3.31}{\pi} \times 10^{-12}}$

$= 5 \times 10^{-16}$J

2. $T_B = \frac{a}{Rb} = \frac{3.6}{0.08 \times 0.6} = 75k$

$\frac{75}{15} = 5K$

3. NaCN, Na_3PO_4, Na_2CO_3, $NaHCO_3$, $Na_2C_2O_4$, Na_2HPO_4

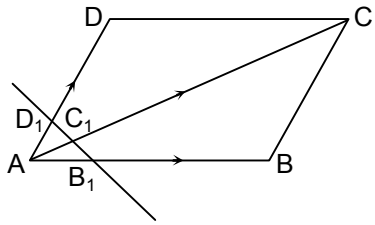
4. $\Delta G^\circ = -2.303\ RT \log K$

5. Fact

Mathematics PART – III

SECTION – A

1. Let $\overrightarrow{AB} = \vec{a}$, $\overrightarrow{AD} = \vec{b}$, then $\overrightarrow{AC} = \vec{a} + \vec{b}$



Given, $\overrightarrow{AB_1} = \lambda_1\vec{a}$, $\overrightarrow{AD_1} = \lambda_2\vec{b}$, $\overrightarrow{AC_1} = \lambda_3(\vec{a} + \vec{b})$

$\overrightarrow{B_1D_1} = \overrightarrow{AD_1} - \overrightarrow{AB_1} = \lambda_2\vec{b} - \lambda_1\vec{a}$

Since, vectors $\overrightarrow{D_1C_1}$ and $\overrightarrow{B_1D_1}$ are collinear

$\overrightarrow{D_1C_1} = k\overrightarrow{B_1D_1}$

$\Rightarrow \lambda_3(\vec{a} + \vec{b}) - \lambda_2\vec{b} = k(\lambda_2\vec{b} - \lambda_1\vec{a})$

$\Rightarrow \lambda_3 = -k\lambda_1$ and $\lambda_3 - \lambda_2 = k\lambda_2$

$\Rightarrow k = \dfrac{-\lambda_3}{\lambda_1} = \dfrac{\lambda_3 - \lambda_2}{\lambda_2} \Rightarrow \lambda_1\lambda_2 = \lambda_1\lambda_3 + \lambda_2\lambda_3$

$\Rightarrow \dfrac{1}{\lambda_3} = \dfrac{1}{\lambda_1} + \dfrac{1}{\lambda_2}$

2. $\det\det(f(\theta+\phi)) = \begin{vmatrix} \cos(\theta+\phi) & -\sin(\theta+\phi) & 0 \\ \sin(\theta+\phi) & \cos(\theta+\phi) & 0 \\ 0 & 0 & 1 \end{vmatrix} = \cos^2(\theta+\phi) + \sin^2(\theta+\phi) = 1$

3. Given $\vec{p} = \vec{a} + \vec{b} - 2\vec{c}$

$\vec{q} = 3\vec{a} - 2\vec{b} + \vec{c}$

and $\vec{r} = \vec{a} - 4\vec{b} + 2\vec{c}$

Given $v_1 = \left[\vec{a}\vec{b}\vec{c}\right]$ ..... (1)

$v_2 = \left[\vec{p}\vec{q}\vec{r}\right] = \begin{vmatrix} 1 & 1 & -2 \\ 3 & -2 & 1 \\ 1 & -4 & 2 \end{vmatrix}\left[\vec{a}\vec{b}\vec{c}\right]$

$v_2 = 15\left[\vec{a}\vec{b}\vec{c}\right]$ ..... (2)

From (1) and (2) $v_2 : v_1 = 15 : 1$

4. $\dfrac{1}{2}\begin{vmatrix} 1 & 5 & 1 \\ 7 & -2 & 1 \\ \alpha & \beta & 1 \end{vmatrix} = 2$ where (α, β) divides PQ in K : 1

5. $\dfrac{\cos x \tan x}{k^2} + \dfrac{1}{\tan x} + \dfrac{\sin x}{1+\cos x} = \dfrac{\sin x}{k^2} + \dfrac{\cos x(1+\cos x) + \sin^2 x}{\sin x(1+\cos x)} = \dfrac{a}{k} + \dfrac{1}{\sin x} = \dfrac{a}{k} + \dfrac{1}{ak}$

6. Let θ be the angle opposite to side c of ΔABC

$\therefore c^2 = a^2 + b^2 - 2ab\cos\theta = (a-b)^2 + 2ab(1-\cos\theta)$

Also, $\Delta = \dfrac{1}{2}ab\sin\theta \Rightarrow 2ab = \dfrac{4\Delta}{\sin\theta}$

$\Rightarrow c^2 = (a-b)^2 + 4\Delta\frac{1-\cos\theta}{\sin\theta} = (a-b)^2 + 4\Delta\tan\frac{\theta}{2}$

For c to be minimum, a = b

$\therefore$ $2ab = 2a^2 = \frac{4\Delta}{\sin\theta}$

$\therefore$ $a = b = \sqrt{\frac{2\Delta}{\sin\theta}}$

7. Here, $\lim\limits_{x\to 0^+} f(x) = \lim\limits_{x\to 0^+} \frac{\tan^2 x}{x} = 1$

and $\lim\limits_{x\to 0^-} f(x) = \sqrt{\cot 1}$ $\left(\because \lim\limits_{x\to 0^-}\{x\} = -1\right)$

8. Take log on both sides and $\log_3 x = t$

$(t-1)\left(t-\frac{1}{2}\right)(t-3) = 0$, $x = \sqrt{3}$, 3, 27

9. $\sin^2 x - \cos^2 x = -\cos 2x \le 1$

$\sqrt{\frac{6}{5}}\left(\frac{1}{\sqrt{2}}\sin x + \frac{1}{\sqrt{3}}\cos x\right) = \frac{\sqrt{3}}{\sqrt{5}}\sin x + \frac{\sqrt{2}}{\sqrt{5}}\cos x$ = sin x · sin ϕ + cos x cos ϕ

Where $\sin\phi = \sqrt{\frac{3}{5}}$, $\cos\phi = \sqrt{\frac{2}{5}}$ = $\cos(x-\phi) \le 1$

$\cos^6 x + \sin^6 x = (\cos^2 x)^3 + (\sin^2 x)^3$ = 1 – 3 sin^2 x cos^2 x = $1 - \frac{3}{4}(\sin 2x)^2 \le 1$

$\cos^2 x + \sin^4 x = 1 - \frac{(\sin 2x)^2}{4} \le 1$

10. $f(x) \le 0$ and $F'(x) = f(x)$

$\Rightarrow f(x) \ge cF(x) \Rightarrow \frac{d}{dx}\left(e^{-cx}.F(x)\right) \ge 0$

$\Rightarrow e^{-cs} F(x)$ is an increasing function

$\Rightarrow e^{-cs} F(x) \ge e^{-c(0)} F(0)$

$\Rightarrow F(x) \ge 0 \Rightarrow f(x) \ge 0$

$\Rightarrow f(x) = 0$

Also, $\frac{d}{dx}\left(e^{-x}\, g(x)\right) < 0$

$\Rightarrow e^{-x} g(x)$ is a decreasing function

$\Rightarrow e^{-x} g(x) < e^{-(0)} g(0)$

$\Rightarrow g(x) < 0$ (as g(0) = 0)

Hence, f(x) = g(x) has one solution x = 0

11. $|x^2 + x - 6| = f(x) + g(x) = g(x)$

$\Rightarrow$ No solution

12.-14. Total number of words

(1) 3 like, 1 different = 20

(2) 2 like, 2 like = 6

(3) 2 like, 2 different = 240

(4) all different = 360

SECTION – B

1. (A) $|\cot x| = \cot x + \frac{1}{\sin x}$

If $0 < x < \frac{\pi}{2} \Rightarrow \cot x > 0$

So, $\cot x = \cot x + \frac{1}{\sin x} \Rightarrow \frac{1}{\sin x} = 0$ (no solution)

If $\frac{\pi}{2} < \cot x < \pi$, $-\cot x = \cot x + \frac{1}{\sin x}$

$\frac{2\cos x}{\sin x} + \frac{1}{\sin x} = 0$

$1 + 2\cos x = 0$ and $\sin x \neq 0 \Rightarrow x = \frac{2\pi}{3}$

(C) $\sin^2\alpha + \sin\left(\frac{\pi}{3} - \alpha\right)\cdot\sin\left(\frac{\pi}{3} + \alpha\right) = \sin^2\alpha + \sin^2\frac{\pi}{3} - \sin^2\alpha = \frac{3}{4}$

(D) $\tan\theta = 3\tan\phi$

$\tan(\theta - \phi) = \frac{\tan\theta - \tan\phi}{1 + \tan\theta\tan\phi} = \frac{2\tan\phi}{1 + 3\tan^2\phi} = \frac{2}{\cot\phi + 3\tan\phi}$

Max if $\tan\phi > 0$, $\frac{\cot\phi + 3\tan\phi}{2} \geq \sqrt{3}$ (using AM $\geq$ GM)

$\Rightarrow (\cot\phi + 3\tan\phi)^2 \geq 12 \Rightarrow \tan^2(\theta - \phi) \leq \frac{1}{3}$

2. (A) $\int \frac{x^4 - 1}{x^3\sqrt{x^4 + x^2 + 1}/x}dx$, Let $\frac{\sqrt{x^4 + x^2 + 1}}{x} = t$

$\int \frac{x^2\cdot\sqrt{x^4 + x^2 + 1}}{x^3\cdot t}dt$, $\left(\frac{x^4 - 1}{x^2\cdot\sqrt{x^4 + x^2 + 1}}\right)dx = dt$

$\int \frac{\sqrt{x^4 + x^2 + 1}}{x\cdot t}dt$, $(x^4 - 1)dx = x^2\cdot\sqrt{x^4 + x^2 + 1}\,dt$

$\int \frac{t}{t}dt$, $\int dt = t + C = \frac{\sqrt{x^4 + x^2 + 1}}{x} + C$

(B) $\int \frac{(x^2 - 1)}{x\sqrt{1 + x^4}}dx$, $\int \frac{(x^2 - 1)}{x^2\sqrt{x^2 + \frac{1}{x^2}}}dx$, Let $x + \frac{1}{x} = t$, $\left(1 - \frac{1}{x^2}\right)dx = dt$

$\int \frac{dt}{\sqrt{t^2 - 2}} = \ln\left(t + \sqrt{t^2 - 2}\right) + C = \ln\left(x + \frac{1}{x} + \sqrt{\left(x + \frac{1}{x}\right)^2 - 2}\right) + C$

$\ln\left(\frac{(x^2 + 1) + \sqrt{1 + x^4}}{x}\right) + C$

(C) $\int \frac{(1+x^2)}{(1-x^2)\sqrt{1+x^4}}dx = \int \frac{\left(1+\frac{1}{x^2}\right)dx}{x\left(1-\frac{1}{x^2}\right)\sqrt{\frac{1}{x^2}+x^2}} = \int \frac{\left(1+\frac{1}{x^2}\right)dx}{\left(x-\frac{1}{x}\right)\sqrt{\left(x-\frac{1}{x}\right)^2+2}}$

Let $\left(x-\frac{1}{x}\right)=t$, $\left(1+\frac{1}{x^2}\right)dx=dt$, $\int \frac{dt}{t\sqrt{t^2+2}}$

Let $t^2 + 2 = y^2$, $2t\,dt = 2y\,dy$

$\int \frac{dy}{t^2} = \int \frac{dy}{(y^2-2)} = \frac{1}{2\cdot\sqrt{2}}\ln\left(\frac{y-\sqrt{2}}{y+\sqrt{2}}\right)+C = \frac{1}{2\sqrt{2}}\ln\left(\frac{\sqrt{t^2+2}-\sqrt{2}}{\sqrt{t^2+2}+\sqrt{2}}\right)+C$

$= \frac{1}{2\sqrt{2}}\ln\left(\frac{\left(\sqrt{t^2+2}-\sqrt{2}\right)^2}{t^2}\right)+C = \frac{1}{\sqrt{2}}\ln\left(\frac{\sqrt{t^2+2}-\sqrt{2}}{t}\right)+C$

$= \frac{1}{\sqrt{2}}\ln\left(\frac{\sqrt{\left(x-\frac{1}{x}\right)^2+2}-\sqrt{2}}{\left(x-\frac{1}{x}\right)}\right)+C = \frac{1}{\sqrt{2}}\ln\left(\frac{\sqrt{x^4+1}-\sqrt{2}x}{(x^2-1)}\right)+C$

(D) $\int \frac{dx}{(1+x^4)\sqrt{\sqrt{1+x^4}-x^2}} \Rightarrow \int \frac{dx}{x^5\left(1+\frac{1}{x^4}\right)\sqrt{\sqrt{1+\frac{1}{x^4}}-1}}$

Let $\sqrt{1+\frac{1}{x^4}}-1=t^2 \Rightarrow \frac{1}{2\sqrt{1+\frac{1}{x^4}}}\left(\frac{-4}{x^5}\right)dx = 2tdt \Rightarrow \frac{dx}{x^5} = -t\sqrt{1+\frac{1}{x^4}}\,dt$

$\Rightarrow \int \frac{-t\sqrt{1+\frac{1}{x^4}}dt}{\left(1+\frac{1}{x^4}\right)t} \Rightarrow -\int \frac{dt}{\sqrt{1+\frac{1}{x^4}}} \Rightarrow -\int \frac{dt}{(t^2+1)} \Rightarrow -\tan^{-1}t+C \Rightarrow C-\tan^{-1}\left(\sqrt{\sqrt{1+\frac{1}{x^4}}-1}\right)$

SECTION – C

1. On substituting $x = \frac{4}{9}$, we get $0 < a < 1$

$\Rightarrow \log_a(x^2 - x + 2) > \log_a(-x^2 + 2x + 3) \equiv 0 < x^2 - x + 2 < -x^2 + 2x + 3$

$\Rightarrow \frac{3-\sqrt{17}}{4} < x < \frac{3+\sqrt{17}}{4}$

2. $\left[\vec{u}\ \ \vec{v}\ \ \vec{w}\right] = \vec{u}\cdot\left(\vec{v}\times\vec{w}\right) = \left|\vec{u}\right|\left|\vec{v}\times\vec{w}\right|\cos\theta$

For maximum value, $\theta = 0^\circ$

$\therefore\ p = \left|\vec{v}\times\vec{w}\right| = \sqrt{59}$

3. $I = \int e^x\left(\frac{1}{(1-x)\sqrt{1-x^2}} + \sqrt{\frac{1+x}{1-x}}\right)dx$

Now, $\frac{d}{dx}\left(\sqrt{\frac{1+x}{1-x}}\right) = \frac{1}{(1-x)\sqrt{1-x^2}}$

$\Rightarrow I = e^x\left(\frac{1+x}{1-x}\right)^{1/2} + c$

$\Rightarrow \mu = 1,\ \lambda = \frac{1}{2}$

4. LHS $= \begin{vmatrix} 2ka\cos A & kc & kb \\ kc & 2kb\cos B & ka \\ kb & ka & 2kc\cos C \end{vmatrix}$ (From sine rule)

$= k^3\begin{vmatrix} 2a\cos A & c & b \\ c & 2b\cos B & a \\ b & a & 2c\cos C \end{vmatrix}$

$= k^3\begin{vmatrix} \cos A & a & 0 \\ \cos B & b & 0 \\ \cos C & c & 0 \end{vmatrix}\begin{vmatrix} a & \cos A & 0 \\ b & \cos B & 0 \\ c & \cos C & 0 \end{vmatrix} = 0$

$\Rightarrow \lambda = 0$

5. Here, $\log_{|x-1|}\left|x^2 - 1\right| = 1$

$\Rightarrow \left|x^2 - 1\right| = \left|x - 1\right|$

$\Rightarrow x = 1, 0, -2$ but $x \neq 0, 1$

$\Rightarrow$ Number of solutions is one